

B.Tech.(CSDA), Sixth Semester

END-SEMESTER EXAMINATION, April-May- 2025

Time- 3hrs

Max. Marks- 40

Course Code: CBCPC19

Course Title- Soft Computing

Note: - Attempt all questions. Missing data/ information if any, may suitably be assumed & mentioned in the answer.

No	Question	Marks	CO
Q1	Attempt any two parts of the following		
1a	Name and explain about the constituents of Soft Computing.	4	I
1b	Discuss the limitations of hard computing? How soft computing can overcome these limitations?	4	I
1c	Consider the universal sets given by: $X = \{(a1, 0.2), (a2, 0.5), (a3, 0.8)\}$ $Y = \{(b1, 0.1), (b2, 0.6), (b3, 0.9)\}$ $Z = \{(c1, 0.4), (c2, 0.7), (c3, 1)\}$, Let the relations R, S, T be defined as: R is subset of $X \times Y$, S is subset of $Y \times Z$ and $T = R \circ S$ (Max-Min Composition). Find T.	4	I
Q2	Attempt any two parts of the following		
2a	Differentiate between Fuzzy Set Theory and Probability Theory.	4	II
2b	Discuss Support and crossover points of a fuzzy Set.	4	II
2c	Compare soft computing with hard computing.	4	II
Q3	Attempt any two parts of the following		
3a	Following are some statements- a. Smita is tall b. Person is fit as per Body Mass Index. c. Chinese are not very tall. d. Generally, boys below 5'5" are called short. Define fuzzy set which can be used to represent the list of above statement.	4	III
3b	Compare Degree, Betweenness centrality measures for fuzzy graph.	4	III
3c	If A and B are two fuzzy sets with membership functions $\mu_A(x) = \{0.6, 0.5, 0.1, 0.7, 0.8\}$ $\mu_B(x) = \{0.9, 0.2, 0.6, 0.8, 0.5\}$ then find the value of disjunctive sum, bounded difference and cartesian product of the sets. Also verify if De Morgan's Law holds on these fuzzy sets or not.	4	III
Q4	Attempt any two parts of the following		

4a	Explain Fuzzy Hypernymy/ Hyponymy Relation, Fuzzy Meronymy/Holonymy Fuzzy attributes, and Fuzzy Function Relations in Fuzzy Hindi WordNet	4	IV																																																						
4b	Discuss Global Fuzzy Graph Connectivity Measures and their importance in problem solving.	4	IV																																																						
4c	Explain how alpha cut can be used for defuzzification.	4	IV																																																						
Q5	Attempt any two parts of the following																																																								
5a	Find Reducts, Core, sample space after reduction and create rules for decision system	4	V																																																						
<table border="1"> <thead> <tr> <th>#</th> <th>Degree</th> <th>CGPA</th> <th>Backlog</th> <th>Recommendation</th> <th>Decision</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>B.Tech.</td> <td>Average</td> <td>No</td> <td>High</td> <td>Granted</td> </tr> <tr> <td>2</td> <td>B.Tech.</td> <td>Fresher</td> <td>No</td> <td>None</td> <td>Not Granted</td> </tr> <tr> <td>3</td> <td>BS</td> <td>Low</td> <td>No</td> <td>Moderate</td> <td>Not Granted</td> </tr> <tr> <td>4</td> <td>MS</td> <td>High</td> <td>No</td> <td>None</td> <td>Granted</td> </tr> <tr> <td>5</td> <td>MS</td> <td>Average</td> <td>No</td> <td>None</td> <td>Not Granted</td> </tr> <tr> <td>6</td> <td>MS</td> <td>High</td> <td>No</td> <td>High</td> <td>Granted</td> </tr> <tr> <td>7</td> <td>B.Tech.</td> <td>High</td> <td>Yes</td> <td>Moderate</td> <td>Granted</td> </tr> <tr> <td>8</td> <td>BS</td> <td>Low</td> <td>Yes</td> <td>High</td> <td>Not Granted</td> </tr> </tbody> </table>				#	Degree	CGPA	Backlog	Recommendation	Decision	1	B.Tech.	Average	No	High	Granted	2	B.Tech.	Fresher	No	None	Not Granted	3	BS	Low	No	Moderate	Not Granted	4	MS	High	No	None	Granted	5	MS	Average	No	None	Not Granted	6	MS	High	No	High	Granted	7	B.Tech.	High	Yes	Moderate	Granted	8	BS	Low	Yes	High	Not Granted
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5b	Is genetic algorithm admissible? Do they guarantee convergence? Explain why a mutation operator is used in a genetic algorithm.	4	V																																																						
5c	Explain Particle Swarm Optimization with Flow Chart. Also discuss its advantages and limitations.	4	V																																																						